

DATA CLEANING REPORT

By Christian Udochukwu Onwuka

Introduction

This project focuses on cleaning and preparing a product dataset containing over 3847 rows and 6 columns using Microsoft Excel. The main objective was to improve data quality and create a short title column that makes product titles more suitable for analysis and SEO purpose.

The dataset contains columns such as Product Id, Title, Bullet Points, Description, Product Type Id, and Product Length.

The **title** column was identified as the most important variable for creating the short title column, while **product ID** was used as a unique identifier to identify and remove duplicate records.

Initial review also showed that column names required standardization for clarity and consistency.

Data Cleaning Process

The dataset contains over 3847 rows and 6 columns

During the initial exploration, several data quality issues were identified:

- Duplicate records were present
- Product titles were long and unstructured
- Inconsistent formatting across text fields
- Presence of unnecessary symbols such as “|”
- Some columns contained HTML-like text and encoding issues

Column names were standardized for consistency. Text fields were cleaned using Excel functions such as **Trim** to remove extra spaces and **Substitute** to remove unwanted characters and symbols like “|”.

In addition, Find and Replace was used to quickly remove or replace repetitive and unnecessary words across the dataset, improving overall readability and consistency. The Proper function was applied to standardize capitalization and enhance readability.

Short Title Creation

The short title feature was created to provide a readable version of the title while retaining essential information.

Product titles were analyzed to identify key components such as product type, attributes, and important keywords. Repetitive and unnecessary words were removed using a combination of **Find and Replace** and Excel formulas.

Excel functions such as Trim, Substitute, Left, and Proper were combined into a single formula to clean, standardize, and shorten the titles. The Proper function ensured consistent capitalization for improved readability. Titles were limited to 50 characters to maintain clarity and meet the required constraints.

Clean Dataset Overview

The final dataset is cleaner, more consistent, and easier to analyze.

Key improvements include:

- Removal of **306 duplicate records**
- Reduction of noise in text fields
- Standardized formatting across columns
- Creation of a **short title** column for improved readability